



East West University
Department of Computer Science and Engineering
Course Outline
Spring 2026

Course Information

Course: CSE251 Electronic Circuits

Credits and Teaching Scheme

	Theory	Laboratory	Total
Credits	3	1	4
Contact Hours	3 Hours/Week for 13 Weeks + Final Exam in the 14 th Week	2 Hours/Week for 13 Weeks	5 Hours/Week for 13 Weeks + Final Exam in the 14 th Week

Prerequisite

CSE209 Electrical Circuits

Instructor Information

Instructor: Rashedul Amin Tuhin [RDA], Senior Lecturer, Department of CSE
Office: Room#640
Tel. No.: 09666775577 (hunting) ext. 447, 01719 356 588 (WhatsApp preferred)
E-mail: mcctuhin@ewubd.edu

Class Routine and Office Hour

Office Hours and Class Routine for Spring 2026					
<i>Ramadan Time</i>	<i>09:00 - 10:20</i>	<i>10:30 - 11:50</i>	<i>12:00 - 13:20</i>	<i>13:30 - 14:50</i>	<i>15:00 - 16:20</i>
Regular Time	08:30 - 10:00	10:10 - 11:40	11:50 - 13:20	13:30 - 15:00	15:10 - 16:40
Sunday (S)	CSE407 (2) AB3#602	Office Hour	Office Hour	CSE251 (5) ICE213 (5) AB2#702	CSE251 (6) ICE213 (6) AB2#601
Monday (M)	Office Hour				
Tuesday (T)	Office Hour	CSE407 (1) AB3-801	Office Hour	CSE251 Lab (6) ICE213 Lab (6) R#548 (Electrical Circuit Lab)	
Wednesday (W)	Office Hour	CSE251 Lab (5) ICE213 Lab (5) R#548 (Electrical Circuit Lab)		Office Hour	Office Hour
Thursday (R)	CSE407 (2) AB3#602	CSE407 (1) AB3#801		CSE251 (5) ICE213 (5) AB2#702	CSE251 (6) ICE213 (6) AB2#601

Course Objective

The subject aims to provide students with the ability to use fundamental electrical and electronic abstractions to analyze and design circuits and systems built with lumped and electronic circuit

elements. This course provides basic knowledge of how complex devices such as semiconductor diodes, operational amplifiers (op-amps), and bipolar and field-effect transistors are modeled and used to design and analyze practical circuits. Besides, this course also emphasizes the practical implementation of building, testing, and performance analysis of electronic circuits. Knowledge of this course will be a prerequisite for future courses such as CSE345 Digital Logic Design, CSE350 Data Communications, CSE360 Computer Architecture, CSE442 Microprocessor and Microcontrollers, and CSE490 VLSI Design.

Knowledge Profile

K1: Theory-based natural sciences

K3: Theory-based engineering fundamentals

Learning Domains

Cognitive - C2: Understanding, C3: Applying,

Psychomotor - P2: Manipulation, P3: Precision

Affective - A2: Responding

Program Outcomes (POs)

PO1: Engineering Knowledge

PO2: Problem Analysis

Complex Engineering Problem Solution

EP1: Depth of knowledge required

EP2: Range of conflicting requirements

Complex Engineering Activities

None

Course Outcomes (COs) with Mappings

After completion of this course, students will be able to:

CO	CO Description	PO	Learning Domains	Knowledge Profile	EP/EA
CO1	Understand and use the fundamental concepts of the diode, Bipolar Junction Transistor (BJT), MOSFET, and Operational Amplifier (Op-amp) for solving electronic circuits.	PO1	C2, C3	K1	
CO2	Use and justify the Diode, BJT, and MOSFET for designing electronic circuits.	PO2	C3	K2	
CO3	Use and justify Operational amplifiers for designing electronic circuits.	PO2	C3	K2	
CO4	Use analytical, software, and hardware techniques, perform and demonstrate skills, and write reports to design, build and test electronic circuits.	PO1	C3 P2, P3 A2	K1, K2	EP1, EP2

Course Topics, Teaching-Learning Method, and Assessment Scheme

Lecture Plan (L)	Course Topic	Subtopics/Details	Teaching-Learning Method	CO	Mark of Cognitive Learning Levels			Mark of COs	Exam Mark
					C 2	C 3	C 4		
L1	Semiconductor diodes	Semiconductor materials, Intrinsic materials, Extrinsic materials, n-type and p-type materials. Majority and minority carriers.	Lecture, Slides, Class Discussion, Discussion with Instructor/TA	CO1	3	4		7	Mid-Semester Assessment (30)
L2	Semiconductor diodes	Operation and characteristics of semiconductor diodes; forward and reverse bias conditions; load-line analysis.							
L3	Applications of the Diode	Series diode configuration, parallel diode configuration, series parallel diode configuration, Rectifier circuits, Clipper and Clamper, Zener Diode.	Do	CO2					
L4	Applications of the Diode	Rectifier circuits (half-wave rectifier and full-wave rectifier circuits), AND/OR gates representation using a diode.							
L5	Applications of the Diode	Clipper and Clamper circuits							

L6	Applications of the Diode	Zener Diode (Introduction to Zener Diodes, Zener Breakdown Mechanism, Zener Diode as a Voltage Regulator, Applications of Zener Diodes)							
L7	Transistor construction	Device Structure and Physical Operation of BJT							
L8 CT1	Transistor Operation and Application of BJT	Transistor construction and operation. Current-Voltage Characteristics, mode of operation	Do	CO1	5			5	
		(Active region, Cut off region, saturation region),							
L9	Transistor Operation and Application of BJT	Common base configuration, common emitter follower configuration, and common collector configuration. Load line analysis.							
L10	BJT circuits at DC, Biasing in BJT amplifier circuits. Application of BJT	BJT as an amplifier and switch, Fixed bias configuration							
L11	BJT circuits at DC, Biasing in BJT amplifier circuits. Application of BJT	Emitter bias configuration, voltage divider bias configuration	Do	CO2			8	8	

L12 CT2	BJT circuits at DC, Biasing in BJT amplifier circuits. Application of BJT	Collector feedback configuration, miscellaneous bias configuration							
	Mid-Semester Assessment								
L13	Operational Amplifier Circuits	Characteristics of the ideal Op-amp, Basic Comparator circuits	Do	CO1	3	4		7	Final Exam (30)
L14	Operational Amplifier Circuits	Inverting and non-inverting comparator, zero crossing detector							
L15	Operational Amplifier Circuits	Inverting and non-inverting amplifiers, Voltage follower							
L16	Application of OPAMP circuits	Applications such as Adder and Difference amplifiers	Do	CO3		3	5	8	
L17	Application of OPAMP circuits	Close loop gain, open loop gain, Difference and Common mode operation							
L18	Application of OPAMP circuits	Integrator and Differentiator, and Instrumentation amplifiers							
L19	MOSFET construction	Device Structure and Physical Operation of MOS, Enhancement type and Depletion type MOS, n- channel and p-channel MOS, Complementary MOS or CMOS	Do	CO1	3	4		7	

L20	MOSFET operations	Modes of Operation, Current-Voltage Characteristics, MOSFET as an Amplifier and Switch						
L21 CT3	MOSFET operations and application	DC biasing (Fixed Bias, Voltage divider bias)						
L22	MOSFET operations and application	DC biasing (Self-bias, Drain-to-gate feedback bias)	Do	CO2	3	5	8	
L23	MOSFET operations	Small signal equivalent models of MOSFET (Voltage gain, Current gain, Input and output resistance, Frequency response)						
L24	MOSFET operations and application	Logic circuit using n-MOS and p-MOS						
Final Exam								

Laboratory Experiments and Assessment Scheme

Experiment	Teaching-Learning Method	CO	Mark of Cognitive Learning Level	Mark of Psychomotor Learning Levels		Mark of Affective Learning Level	CO Mark
			C3	P2	P3	A2	
I-V Characteristics and Modelling of Forward Conduction of a Diode	Preparing Pre-Lab Report, Lab Experiments, and Result Analysis	CO4					

Diode Rectifier Circuits	Do	CO4					
Study of clipper and clamper circuits	DO	CO4					
Study of the Zener Diode	Do	CO4					
Biasing of the BJT Common-Emitter Circuit	Do	CO4					
Adder and Amplifier Circuits Using Op Amp	Do	CO4					
Signal Integration Using Op-Amp	Do	CO4					
Signal Integration Using Op-Amp	Do	CO4					
Measurement of Parameters and I-V characteristics of an N-channel MOSFET	Do	CO4					
Biasing of a Common-Source Voltage Amplifier	Do	CO4					
Total Lab Performance		CO4	4	1	0	0	5
Lab Exam		CO4	7	1	1	1	10
Total			11	2	1	1	15

Mini Project

Mini Project	Teaching-Learning Method	CO	EP/EA	Mark of Cognitive Learning Level	Mark of Psychomotor Learning Levels	Mark of Affective Learning Level	CO Mark
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				C3	P2	P3	A2	
Mini Project including Report and Presentation	Group-based, moderately complex electronic circuit building for practical application, with report writing and presentation	CO4	EP1, EP2	3	0.5	0.5	1	5

Overall Assessment Scheme

Assessment Area	CO					Total	PO Marks	
	CO1	CO2	CO3	CO4	Other		PO1	PO2
Class Test/Quiz	5	5	5			15	5	10
Mid-Semester Assessment	12	18				30	12	18
Final Exam	14	8	8			30	14	16
Laboratory Performance and Lab Exam				15		15	15	
Mini Project				5		5	5	
Class Performance					5	5		
Total	31	31	13	20	5	100	51	44

Teaching Materials

Textbook:

1. Adel S. Sedra and Kenneth C. Smith, *Microelectronic Circuits*, New Delhi, Oxford University Press, 2019.

Reference Book:

1. Charles K. Alexander and Matthew N. O. Sadiku, *Fundamentals of Electrical Circuits*. New Delhi, McGraw-Hill Education, 2020.

Lab Manual: Lab manuals will be provided.

Project Description: Project descriptions will be provided.

Equipment/Software: Digital trainer board, and Computer & PSpice Software.

Grading System

Marks (%)	Letter Grade	Grade Point
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
45% to less than 50%	C	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

Exam Dates

Section	Class Slot	Mid Semester	Final
5	SR	Thursday March 5, 2026	Thursday April 30, 2026
6	SR	Sunday March 8, 2026	Thursday April 30, 2026

Academic Code of Conduct

Academic Integrity:

Any form of cheating, plagiarism, personification, or falsification of a document, as well as any other form of dishonest behavior related to obtaining academic gain or the avoidance of evaluative exercises committed by a student, is an academic offense under the Academic Code of Conduct and **may lead to severe penalties as decided by the Disciplinary Committee of the university.**

Special Instructions:

- Students are expected to attend all classes and examinations. A student **MUST** have at least 80% class attendance to sit for the final exam.
- Students will not be allowed to enter the classroom after 20 minutes of the starting time.
- For plagiarism, the grade will automatically become zero for that exam/assignment.
- Normally there will be **NO make-up exam**. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student misses any exam, the student **MUST** get approval for a makeup exam by written application to the Chairperson through the Course Instructor **within 48 hours** of the exam time. Proper supporting documents in favor of the reason for missing the exam have to be presented with the application.
- For the **final exam**, there will be **NO** makeup exam. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student misses the final exam, the student **MUST** get the approval of an **Incomplete Grade** by written application to the Chairperson through the Course Instructor **within 48 hours** of the final exam time. Proper supporting documents in favour of the reason for missing the final exam have to be presented with the application. **It is the responsibility of the student to arrange an**

Incomplete Exam within the deadline mentioned in the Academic Calendar in consultation with the Course Instructor.

- All mobile phones **MUST** be turned to silent mode during class and exam periods.
- There is **zero tolerance for cheating** in exams. Students caught with cheat sheets in their possession, whether used or not; writing on the palm, back of calculators, chairs, or nearby walls; copying from cheat sheets or other cheat sources; copying from other examinees, etc., would be treated as cheating in the exam hall. The only penalty for cheating is **expulsion for several semesters, as decided by the Disciplinary Committee of the university.**

Course Instructor